# MAGNUS DIERKING

*Born in Germany*, 21 September 1998

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## WORK EXPERIENCE

*08/2025–Present* Research Assistant

*IAS* *Intelligent Autonomous Systems Group · Technical University of Darmstadt*
Developing a Python/C++ software stack for FR3 robotic arms, built on ROS2 Jazzy.

*10/2024–05/2025* Research Intern

*Huawei R&D* *Huawei Technologies Research & Development UK Limited · London, Shenzhen*
Extracurricular internship during Master studies. Working on reinforcement learning, imitation learning and supporting middleware for robotics as part of the Embodied AI team.

*09/2023–09/2024* Research Assistant

*IAS* *Intelligent Autonomous Systems Group · Technical University of Darmstadt*
Assited Ph.D. researchers with experiments, visualization and studies on supporting theory in optimal transport, Bayesian inference and optimization for robot learning.

*2018–2021, 2022–2023* Working Student

*mas* *Medical Airport Service GmbH · Frankfurt*
Support personel for the rescue station on the apron of Frankfurt International Airport.

*2021–2022* Teaching Assistant

*TEMF* *Institute for Accelerator Science and Electromagnetic Fields · Technical University of Darmstadt*
Developed lesson plans, theoretical / programming tutorials and provided mentorship and feedback for undergraduate students.
Courses: Electrodynamics, Numerics for Electromagnetic Field Simulation

## PUBLICATIONS

*2025* Ark

*arXiv* Open-source, Python-first robotics framework that provides a Gym-style interface for collecting data, training policies, and switching seamlessly between simulation and real-robot deployment. It includes reusable modules for control, SLAM, motion planning and visualization, and integrates natively with ROS to accelerate end-to-end robotics research.
*First Author*

*2025* OpenPyro-A1

*IEEE RA-L* Open-source, low-cost bimanual half-humanoid robot designed for advanced manipulation research. It features a modular, repairable hardware design and supports coordinated two-handed tasks such as folding, cutting, and assembling. The platform enables teleoperation via a Meta Quest 3 and provides interfaces for learning-based controllers to support scalable, real-world robotics experimentation.
*Co-Author*

## EDUCATION

*Master of Science*

*10/2022–Present* **Computational Engineering**

*GPA: 1.00* · *Technical University of Darmstadt*
Thesis: *Model Tensor Planning*
Ongoing.
Advisor: Prof. Jan Peters · Supervision: Dr. João Carvalho, Dr. An Thai Le

Robot Learning · Reinforcement Learning · Deep Generative Models · Intelligent Robotic Manipulation · Parallel Programming
Differential & Riemannian Geometry · Numerical Linear Algebra · Geometric Algebra · Information Theory 1-2 · Convex Optimization · Graph Signal Processing · Control Theory · Optimal Transport

*Bachelor of Science*

*10/2019–10/2022* **Computational Engineering**

*GPA: 1.51* · *Top 10%* · *Technical University of Darmstadt*
Thesis: *Parallel Solution of Linear Systems Arising in Domain Decomposition Methods*
C++ implementation of a parallel solver for large-scale surface PDEs within an Isogeometric Analysis (IGA) framework.
Advisor: Prof. Sebastian Schöps · Supervision: MSc. Maximilian Nolte · *Grade: 1.00*

Robotics · Software Engineering · Algorithms and Data Structures · Functional & Object-oriented Programming · Geometric Modelling
Partial Differential Equations · Numerical and Statistical Methods · Mechanics 1-3 · Signals and Systems · Electrodynamics 1-2 · Maths 1-4

*10/2018–10/2019* **Mechanical Engineering**

*Completed Coursework* · *Technical University of Kaiserslautern*

Material Science 1-2 · Material Science Lab · Mechanical Design Project · Production Engineering · Business Organization

*A-Levels*

*2018* **Elisabeth-Langgässer Gymnasium**

Math · English · History

## TECHNICAL SKILLS

*OS*: Linux · Mostly Ubuntu, some Arch

*Programming*:
Python · JAX, PyTorch, CVX, NumPy, SciPy, Pandas
C++ · Eigen, OpenMP
Basics · Java, HTML, JavaScript, CSS

*Simulation*: MuJoCo, PyBullet

*Tools*: Git, ROS2, Docker, LaTeX

*Hardware*: Franka Research 3, Trossen Viper, OpenPyro-A1
OptiTrack, Intel RealSense

## OTHER INFORMATION

*Scholarships*
2024 · Erasmus Placements Program
2023 · Deutschlandstipendium

*Languages*
German · Mothertongue
English · Fluent
French · Basics

February 7, 2026